



Chemistry Ch 4.1: The Periodic Table

Chapter 4, Periodic Table and Periodicity · kitab ke page 45 se 72 · sections 4.1.1 aur 4.1.2

Do simtein, do naam

Periods: aarzi (horizontal) qatarein. Ek period mein chalne ka matlab hai usi shell mein electrons barhte jana.

Groups: amoodi (vertical) column. Ek group ke tamam elements ke **bairooni electrons barabar** hote hain, isi liye un ki **khassosiyat multi julti** hoti hai.

Blocks: aakhri electron kahan gira

Element ka **block** us sub-shell se tay hota hai jis mein **aakhri electron** jata hai.

s block: Group 1 aur 2 (aur helium).

p block: Group 13 se 18.

Yani agar configuration **$3s^1$** par khatam ho to element **s block** ka hai.

Configuration se teen cheezein nikalna

Sodium: $1s^2 2s^2 2p^6 3s^1$

Period = sab se bara shell number = **3**. **Group** = bairooni electrons = **1**. **Block** = aakhri sub-shell = **s**.

Teenon ek hi configuration se nikal aate hain. Yehi sawal imtihan mein aata hai.

Quiz

1. What is the difference between a **period** and a **group**? Why do elements of the same group behave alike?
2. An element has the configuration $1s^2 2s^2 2p^6 3s^2 3p^1$. Give its period, its group and its block, and explain how you worked each one out.